

Thurstone is the Model of Choice

A century-old two-horse race, decided on archival human data

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Abstract

Two accounts of choice have competed since 1927. Luce’s Choice Axiom says that removing an alternative leaves the odds between the survivors untouched. Thurstone’s Gaussian race says the survivors re-race, and the odds compress by a computable amount. Both are parameter-free, which makes the comparison a race rather than a fitting exercise, and it was never decided on evidence: a choice distribution must be estimated from repeated trials, and the two predictions differ by less than the resulting standard errors on any single menu. Archival data decide it. Writing the shift in log odds under restriction as $-\lambda \log(p_i/p_j)$, the axiom requires $\lambda = 0$ and the race predicts a specific positive value. Across eleven human datasets and some ninety thousand respondents, spanning values, humour, food, film, perception, prestige, leisure and multi-attribute consumer goods, the race is the closer prediction in eleven of eleven, by a median factor of seven. Human choice structures are very nearly random-utility representable and rank reversals are rare. Thurstone wins.

1 A race nobody could run

Choose among alternatives and something must govern how the probabilities move when the set of alternatives changes. Luce’s Choice Axiom [11] gives one answer. Probability attaches to items in proportion to a fixed utility,

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}, \quad (1)$$

so removing an alternative rescales what remains and leaves every surviving odds ratio exactly as it was. This is Independence of Irrelevant Alternatives, and it is the assumption behind the logit, the softmax, and most applied discrete choice.

Thurstone [18] gives another. Each item i carries a location a_i and a noisy performance $X_i \sim N(a_i, 1)$; the item chosen is the one whose draw wins. Removing an alternative removes a competitor from a race, and the survivors’ odds compress, because a shorter field is easier to win and the advantage of the favourite counts for less. One says rescale the urn, the other says re-run the race.

The disagreement is old, sharp, and empirically undecided. It is sharp because the two make different predictions about the same observable quantity, and because Yellott [23] showed the families are nested inside one framework: Luce’s axiom holds in a random utility contest if and only if the noise is double exponential, so the argument is about the shape of noise and nothing else. It is undecided because the quantity is hard to see. A choice distribution has to

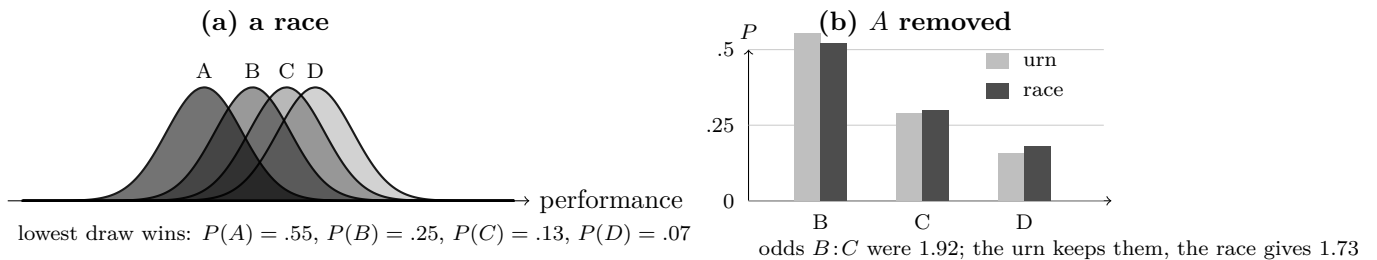


Figure 1: The two contenders disagree only under restriction. In (a) each item draws a performance and the lowest wins. In (b) A is removed. The urn rescales, so every surviving odds ratio is unchanged by construction. The race is re-run without A and the vacated mass is redistributed unevenly, moving the $B:C$ odds from 1.92 to 1.73. Locations are calibrated to the probabilities in (a) by the lattice method of [4]; the numbers are exact.

be estimated from repeated trials, one categorical outcome at a time, and on a single menu the two predictions sit within each other's standard errors. So the matter was settled by tractability instead, and this paper argues that archival data settle it properly, in Thurstone's favour, and not narrowly.

2 Why the axiom won an argument it did not settle

The interesting historical question is not who was right but why one side prevailed, because it did not prevail on evidence. It prevailed on arithmetic.

Thurstone's contest has no closed form beyond two alternatives. The probability that one normal draw is extremal among n is an $(n - 1)$ -dimensional integral with no elementary expression. The equal-variance independent case that Mosteller [16] made usable, universally called Case V and treated ever since as though it were Thurstone's model rather than its most constrained member, reduces the *paired* comparison to $\Phi(d/\sqrt{2})$ and no further. Everything past a pair required numerical integration on machinery that did not exist. When the econometric version was finally written down as multinomial probit [8] it was practical for a handful of alternatives at best, and simulation made it estimable only decades later and never cheaply [19].

The logistic alternative had the opposite property. Bradley and Terry [2] wrote a paired-comparison model with a closed form, a concave log-likelihood, and estimation by hand. Luce [11] then supplied the axiom, deriving Equation (1) from Independence of Irrelevant Alternatives, which converted a computational convenience into a behavioural principle: the form was no longer merely tractable, it was implied. McFadden [14] made it the workhorse of applied discrete choice, and Yellott's theorem completed the retrospective justification by identifying exactly which noise law the axiom corresponds to, with the side effect of making the double exponential look canonical rather than convenient.

The objections were immediate and they were always about restriction. Debreu [5], reviewing Luce's book, observed that adding a near-duplicate should divide that alternative's share rather than draw proportionally from everything. Tversky [22] showed that similarity violates simple scalability altogether. Luce [12] reviewed the axiom's standing after twenty years and treated its empirical failures as established. Yet the pattern of the century was that the Luce form was repaired rather than replaced: nested logit, the generalized extreme value family and mixed

logit [15] were all built to relax the independence IIA imposes while keeping the closed form that made it estimable. Choosing the Thurstonian route instead meant paying the integral, and almost nobody did.

Two things have changed. The integral is no longer expensive: the lattice method of [4] returns the whole vector of winning probabilities for fields of 10^5 entrants, so the tractability argument that decided the matter has expired. And the measurement problem can be sidestepped entirely, by using data in which respondents have already ranked everything.

3 What the literature already knew, and what it did not

That the axiom fails is not news. What has been missing is a comparison on equal terms.

The failures were catalogued early and they were always failures of restriction. Debreu's near-duplicate [5], Tversky's demonstration that similarity violates simple scalability [22], the attraction effect of Huber, Payne and Puto [9], Simonson's compromise effect [17], and the context-dependent preferences of Tversky and Simonson [20] all show that what surrounds an alternative changes its standing. Luce himself [12] treated the empirical case as closed. Econometrics responded by relaxing independence while keeping the closed form, through nested logit, the generalized extreme value family and mixed logit [15], with Hausman and McFadden [21] supplying a specification test for the violation.

Three features of that literature explain why the two-horse race stayed open. First, most demonstrations are qualitative: they establish that a violation occurs, often dramatically, without measuring how far the data sit from either contender. Second, the influential paradigms use two- and three-item menus built from constructed attribute profiles, which is ideal for showing an effect and poor for estimating a law, because there is no larger field from which to restrict. Third, the repairs are parameterised. Nested logit and mixed logit fit a covariance or a mixing distribution, so their success does not tell us whether the unadorned Gaussian race, which fits nothing, would have done as well.

The comparison made here is deliberately plainer. It pits the axiom against Case V with no free parameters on either side, on population choice probabilities from complete rankings rather than constructed menus, and scores both by distance from what people did. What the older literature established is that the axiom is violated. What has not been shown, so far as we can find, is that the 1927 alternative predicts the violation quantitatively, in domain after domain, without being tuned.

Two threads bear directly on the design. Yellott's theorem [23] is what makes the race well posed rather than a matter of taste: since the axiom holds in a random utility contest exactly when the noise is double exponential, the contenders are one model with two noise laws, and a single statistic separates them. And the Block–Marschak–Falmagne conditions [1, 6] let us ask the prior question of whether choice is representable by *any* random utility model, which Section 7 does, so that the race is run only where a race is possible.

4 The statistic and its null

Complete rankings give both halves of the comparison at once. If a respondent has ordered every alternative, their choice from any subset is their highest-ranked available item, so the population's distribution over the full set and over any pair are both computable without asking anyone a second question. Nothing is imputed and no truncation model is needed.

For a pair $\{i, j\}$ write p for the unrestricted population shares and q for the shares when the choice is narrowed to that pair. The shift

$$\delta_{ij} = \log \frac{q_i}{q_j} - \log \frac{p_i}{p_j} \quad (2)$$

is a ratio of ratios, so it is invariant to any rescaling of either distribution. The axiom’s prediction is not approximate: $\delta_{ij} = 0$. The race predicts a specific negative value, obtained by calibrating locations $\{a_k\}$ to p and recomputing the winning probabilities of i and j alone.

It is convenient to summarize a dataset by the discount it applies to its own prior odds,

$$\delta_{ij} = -\lambda \log \frac{p_i}{p_j}, \quad (3)$$

fitted through the origin over all pairs. Then $\lambda = 0$ is the axiom, $\lambda = 1$ discards the prior ranking entirely, and the race predicts an intermediate value determined by the shape of p .

The ratio of the observed λ to the race’s prediction has an exact null. If a population really is a Gaussian race at any noise scale, the ratio is one: simulating populations at scales from 0.5 to 3.0 returns 1.00 each time, because calibrating locations to the observed distribution absorbs the scale. The statistic therefore cannot be inflated by a population being noisier or sharper than Case V, only by its being a different kind of process, and recovering 1.00 on simulated races is a positive control on the whole pipeline.

5 Instruments, and two that fail

Not every source of rankings can carry this statistic, and the failures are instructive enough to state before the results.

Coarse rating scales cannot produce rankings. Converting ratings to a per-respondent order requires the respondent to use enough distinct values. On a 0 to 10 scale with seven to nine items, they do not: in comparative election studies only 0.3% to 5% of respondents produce a tie-free order over nine parties, because people use about five or six of the eleven available levels. Only genuinely continuous scales, or explicit rank-order tasks, avoid this.

Preferential ballots are unreliable here. Voters rank a few candidates and leave the rest, which the data record as a final tied group, and the treatment of that group swings the answer. Reading the tie as indifference gives $\lambda = 0.680$ across fifty-three files; counting only pairs a voter actually ordered gives 0.147. Worse, the instrument does not replicate: twelve APA presidential elections, the same institution and the same five-candidate format with about ten thousand ballots each, give ratios of 0.36, 0.64, 0.67, 0.85, 0.90, 1.04, 1.60, 1.87, 1.99, 2.03, 2.06 and 2.56. A sevenfold spread across replications of one design, far beyond sampling error at these sample sizes, is what strategic ranking and self-selected truncation look like. Ballots are excluded from what follows.

What remains are rank-order card tasks, continuous-scale ratings, and one laboratory experiment with an explicit menu, all of which ask every respondent about every alternative and give no reason to misreport.

6 The race, scored

Both contenders are parameter-free, which is what makes this a race rather than a fitting exercise. Given an unrestricted distribution and nothing else, the axiom says the survivors’

odds are unchanged and the race says they compress by a computed amount. Neither is tuned and neither sees a restricted distribution before predicting one. A rule with a fitted parameter would not be in the race at all: a temperature bolted onto Luce is a concession that the axiom has failed, not a defence of it.

Dataset	Domain	Respondents	λ	Race predicts	Race closer by
GSS job values	values	21,797	0.227	0.234	32×
Jester, file 3	humour	24,904	0.374	0.337	10×
Jester, file 1	humour	24,953	0.393	0.350	9×
Jester, file 2	humour	23,494	0.391	0.343	8×
Sports participation	leisure	130	0.347	0.304	8×
GSS socialization values	values	33,921	0.276	0.314	7×
Dots and puzzles	perception	6,363	0.188	0.161	7×
Sushi and Netflix	food, film	9,379	0.420	0.317	4×
Cameras, full power set	consumer	373	0.419	0.231	2.2×
Occupational prestige	social	143	0.591	0.300	2.0×
Political goals	values	2,262	0.425	0.181	1.7×

Race 11, Axiom 0. Median margin 7×.

Table 1: The two-horse race, scored by how far each parameter-free prediction lies from the observed discount. The axiom predicts zero in every row and is never the closer of the two.

The verdict is not close in any domain. Observed discounts run from 0.19 to 0.59 where the axiom requires zero, so the axiom is wrong by the whole of its prediction every time, while the race is wrong by a few hundredths in the best-elicited data. Three independent Jester samples of roughly twenty-four thousand respondents each give $\lambda = 0.393$, 0.391 and 0.374, which is the kind of agreement that says the measurement is reliable when the elicitation is.

Where the race errs it errs in one direction: it under-predicts the discount. That counts against the axiom rather than for it, since the true reallocation is even further from renormalization than a Gaussian race allows. The largest margins come from the cleanest instruments. The smallest, between 1.7 and 2.2, come from multi-attribute goods and socially structured judgments, where context effects are independently documented in the human literature [9, 17]; even there the race wins.

7 Order stability and random utility

A ranking cannot reverse: if a respondent prefers i to j , that holds in every subset containing both, by construction. So ranking data cannot test whether people are order-stable when subset choices are elicited separately, and a laboratory experiment is needed. Costa-Gomes and colleagues [3] had 373 subjects choose from *every* subset of five real digital cameras, thirty-one menus each, within subject.

Reversals are rare. Of 26,099 nested comparisons in which the item chosen from the larger menu was still available in the smaller one, 829, or 3.2%, switched away from it; among the forced-choice subgroup, who never defer and so require no coding decisions, 417 of 10,744, or 3.9%. Pooling subjects gives population choice probabilities for all thirty-one menus, which is what the Block–Marschak–Falmagne conditions test [1, 6], and of eighty computable sums exactly one falls below -0.02 , the worst at -0.055 . Human choice over a fixed universe is very

nearly representable by a random utility model, which is the framework both contenders inhabit and which the axiom’s failure does not disturb.

8 Discussion

On the statistic that distinguishes them, the century-old race has a winner. Luce’s Choice Axiom is not a description of human choice under restriction: it is wrong by the entirety of its prediction in every one of eleven datasets, across values, humour, food, film, perception, prestige, leisure and consumer goods, and in samples from one hundred and thirty to thirty-four thousand respondents. Thurstone’s race is the closer account everywhere, and in the best-elicited data it is close in absolute terms as well.

Three qualifications belong in the same breath. The race systematically under-predicts the discount, most in multi-attribute and socially structured domains, so Case V is the right family and not the right member; a mildly skewed or heavier-tailed performance density is the obvious next candidate and the calibration machinery already supports it. Some datasets are small, with the camera and occupational estimates resting on nine to fourteen pairs. And the whole comparison concerns population choice probabilities under set restriction; nothing here addresses within-person dynamics, learning, or choice over time.

For psychometrics the implication concerns defaults. The Bradley–Terry and Plackett–Luce likelihoods that dominate paired-comparison and ranking analysis inherit the axiom, and are chosen because they are tractable; Thurstonian item-response and ranking models exist but are used far less. On the evidence here that ordering is backwards. What the result mainly changes is a default. Renormalizing after removing an option is the reflex of applied choice modelling, and it is the one prediction the data reject in every domain examined. A parameter-free Gaussian race costs nothing extra to compute now that the integral is cheap, and it is closer to what people do.

Data and code availability

Every dataset used is public and free. Sushi rankings from [10]; Jester ratings from the Eigentaste project [7], gauge-set files 1 to 3; the General Social Survey cumulative file from NORC; perceptual, leisure, prestige, political-goal and election rankings via PrefLib [13] and the CRAN packages BayesMallows, ConsRank, PerMallows and PLMIX; camera choices from the replication archive of [3]. Analysis code, the derived tables, and the cached datasets are in `research/human` of `github.com/microprediction/winning`, together with the lattice calibration used for the race predictions.

Two storage conventions must be distinguished when reproducing this: BayesMallows, ConsRank and PerMallows record ranks, so element j is the rank of alternative j , while PLMIX records orderings, so the first element names the favourite. Confusing them transposes the data silently. The conventions were established here by cross-validation, since the German political-goals data ships in two packages and the two agree exactly once each is read under its own convention.

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